

POLITECNICO
MILANO 1863

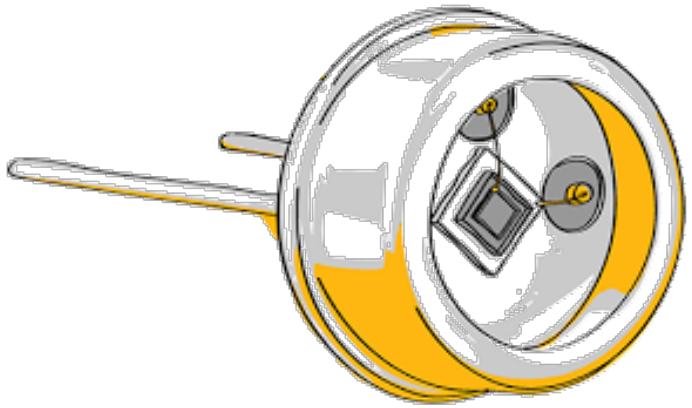
Light sensors

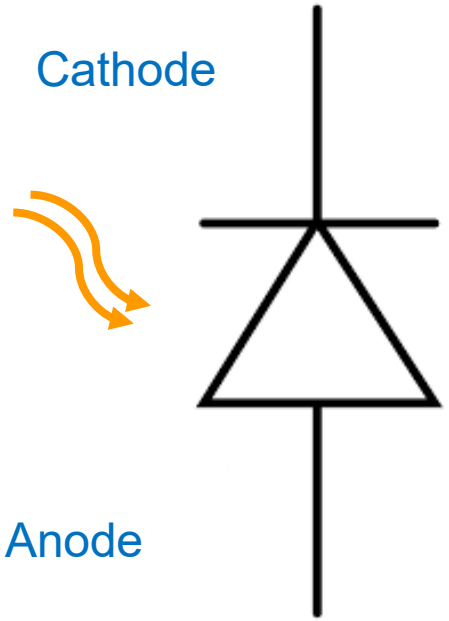
SENSOR SYSTEMS

Federica Villa
federica.villa@polimi.it

- Photodiode
- APD, SPAD and SiPM
- Light Dependent Resistor (LDR)

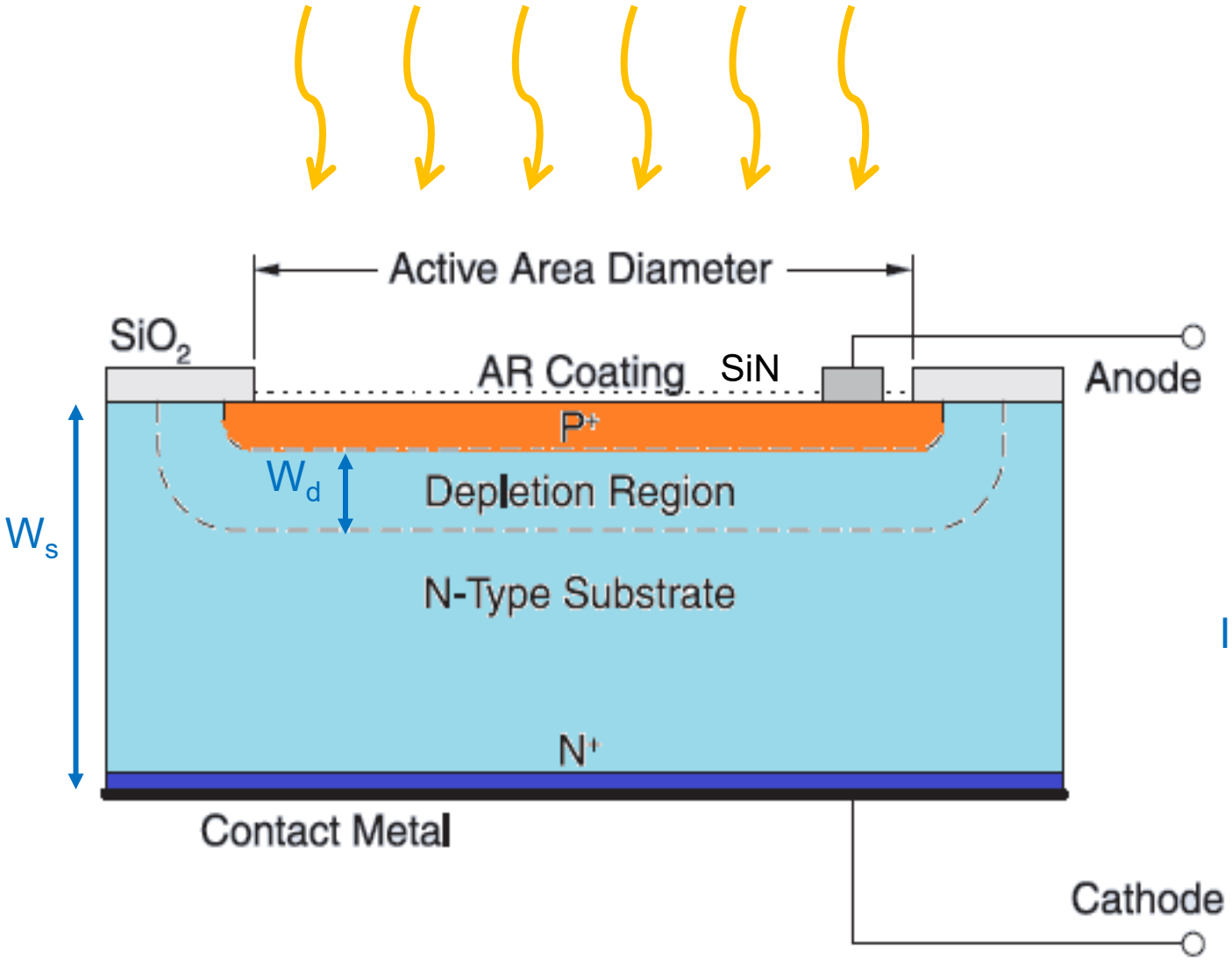






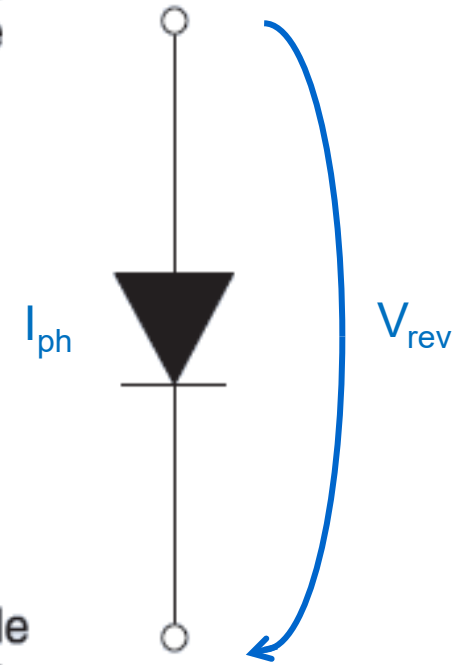
WeBeep “Supplementary materials”: 05 - Photodiode

Silicon Photodiode cross-section



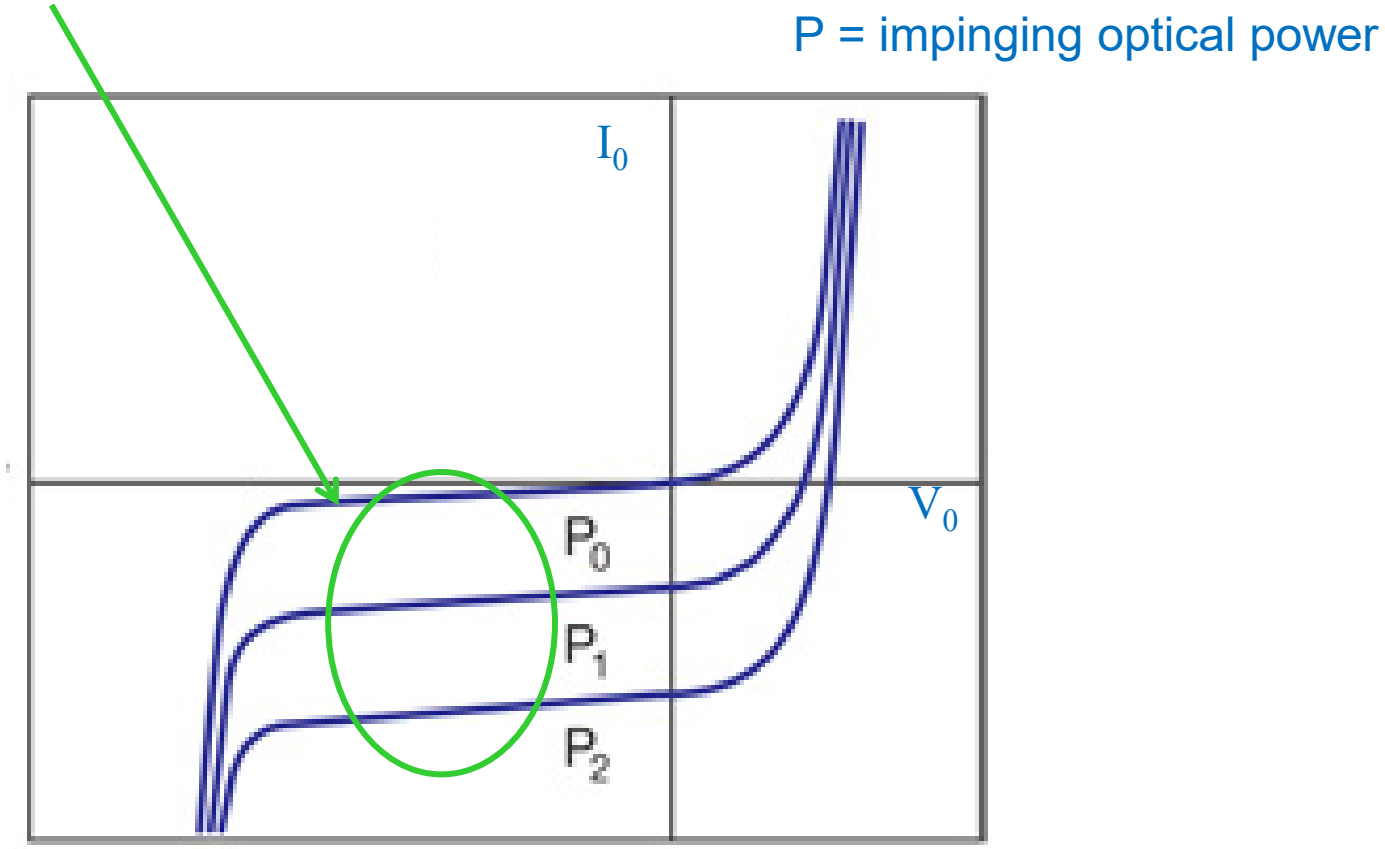
electron-holes:

- photo-generation
- thermal generation

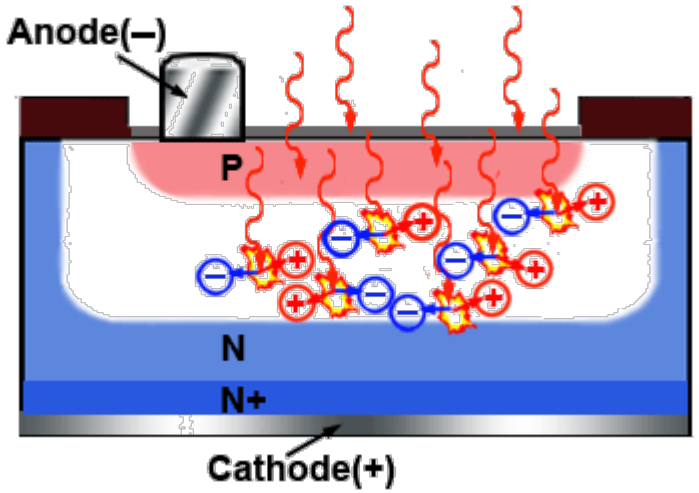


Photoconductive linear mode:

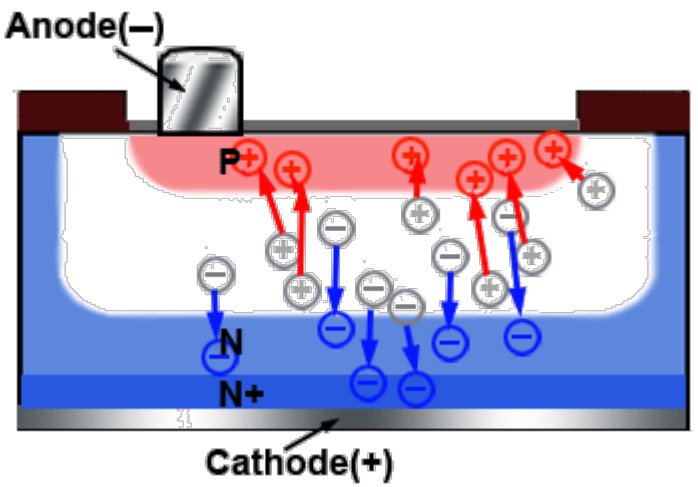
- Reverse bias
- Below break-down voltage



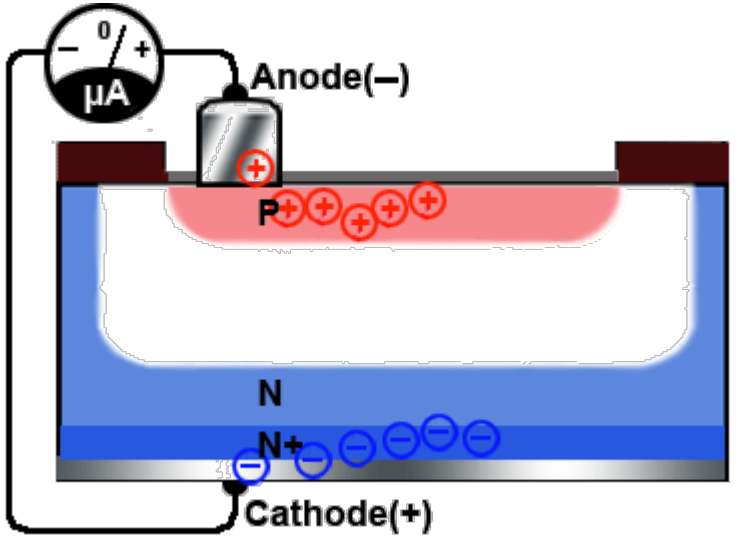
Photodiode Operation



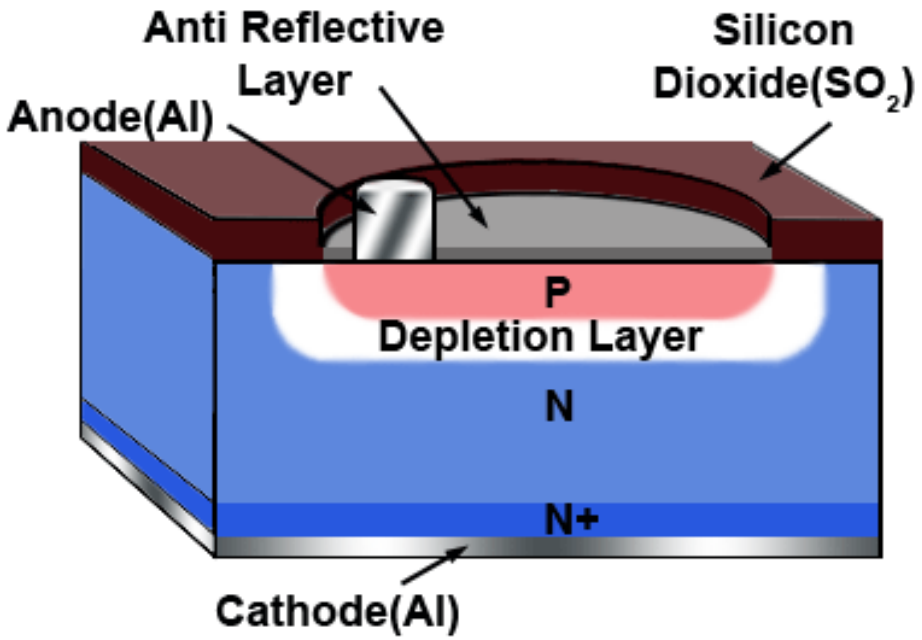
1 - Electron/Hole pairs generation

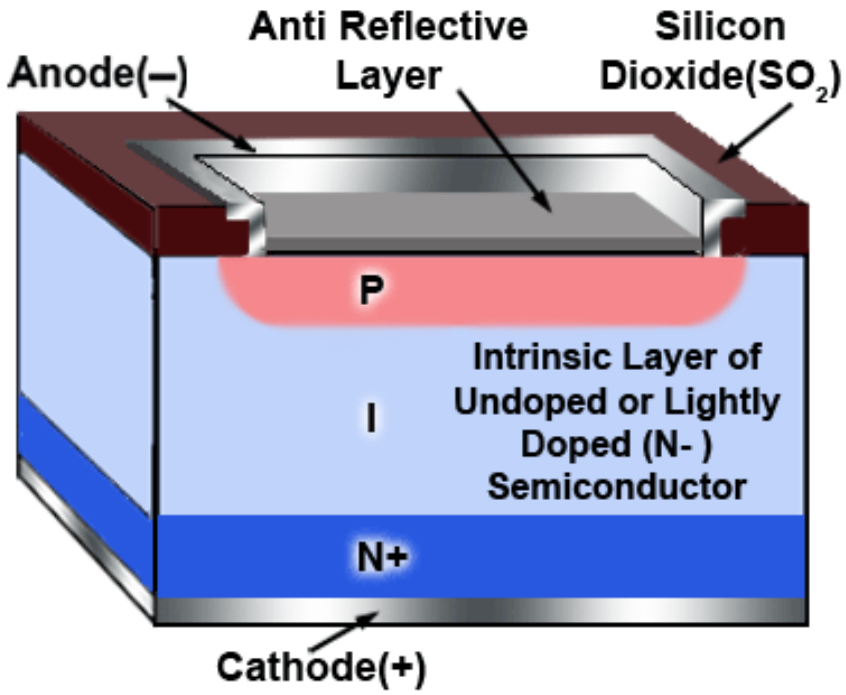


2 - Carriers are attracted by reverse bias



3 - Carriers form a photoelectric current





p-i-n photodiode has larger depletion region

→ reduced capacitance

→ responsivity in a wider spectrum

Penetration depth = depth at which the intensity of the radiation inside the material falls to 1/e (about 37%) of its original value at the surface.

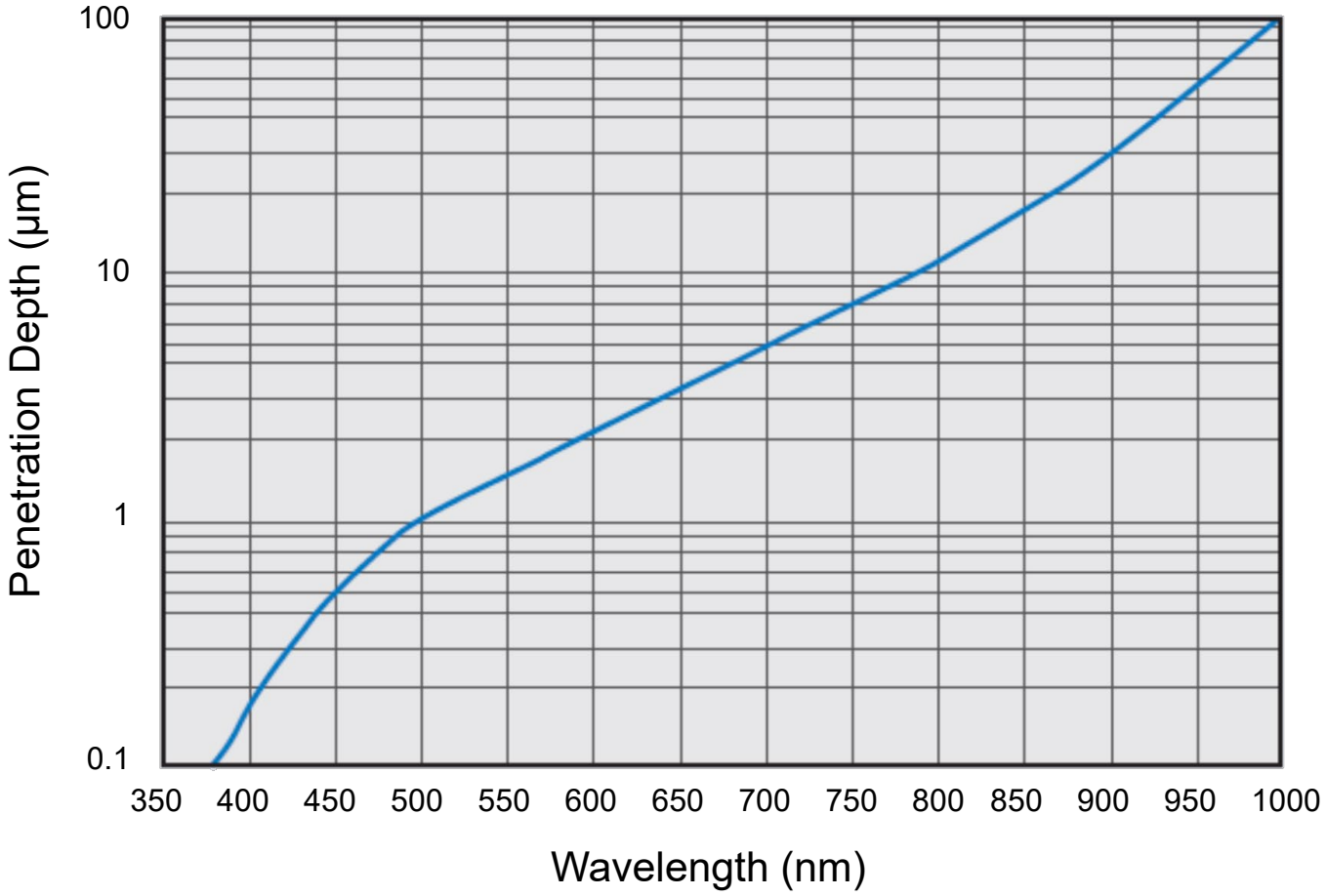
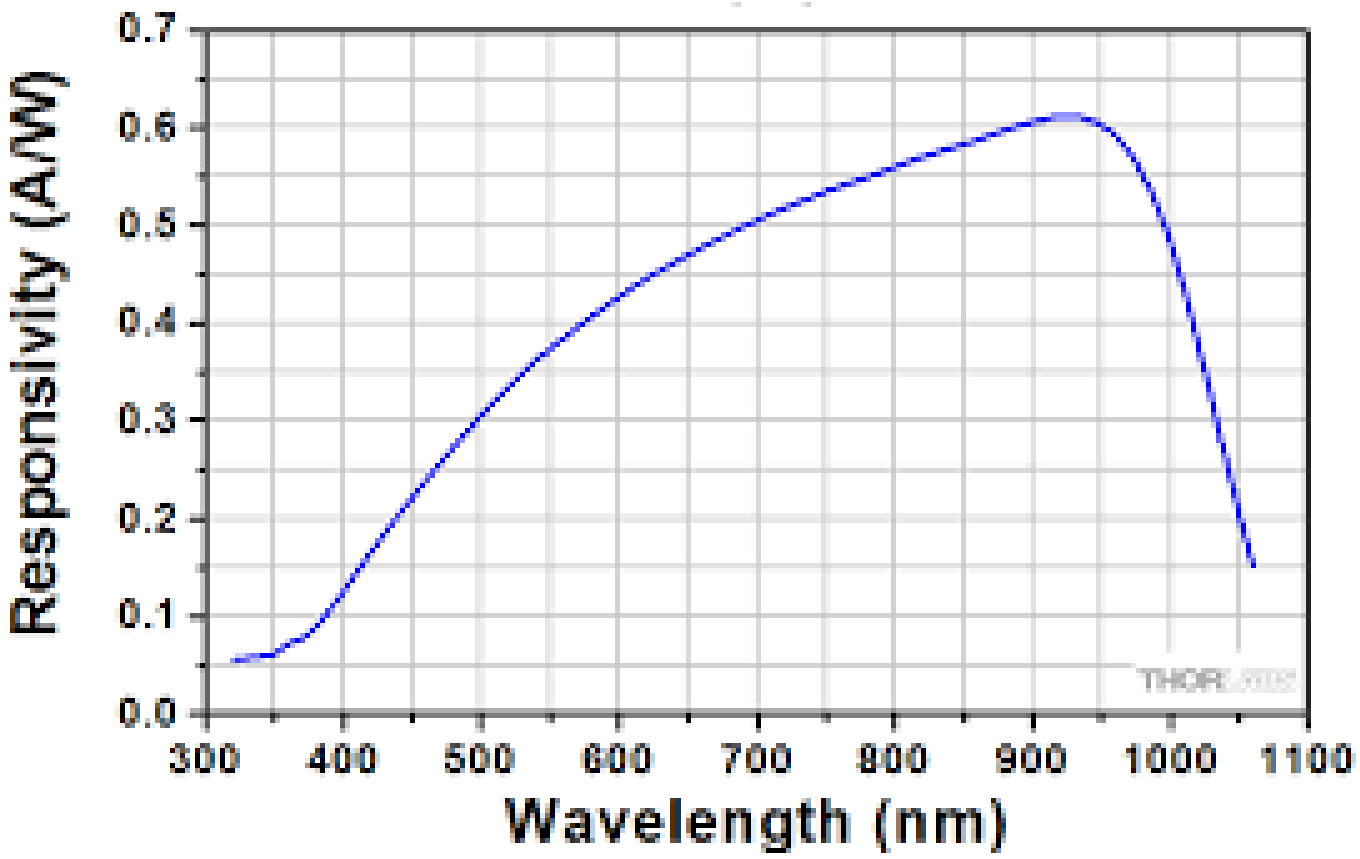


Photo-generation:

Eph > Eg

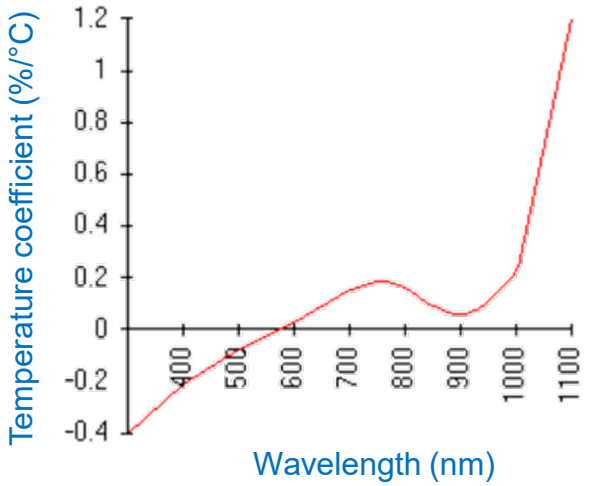
(Si: Eg = 1.12 eV)

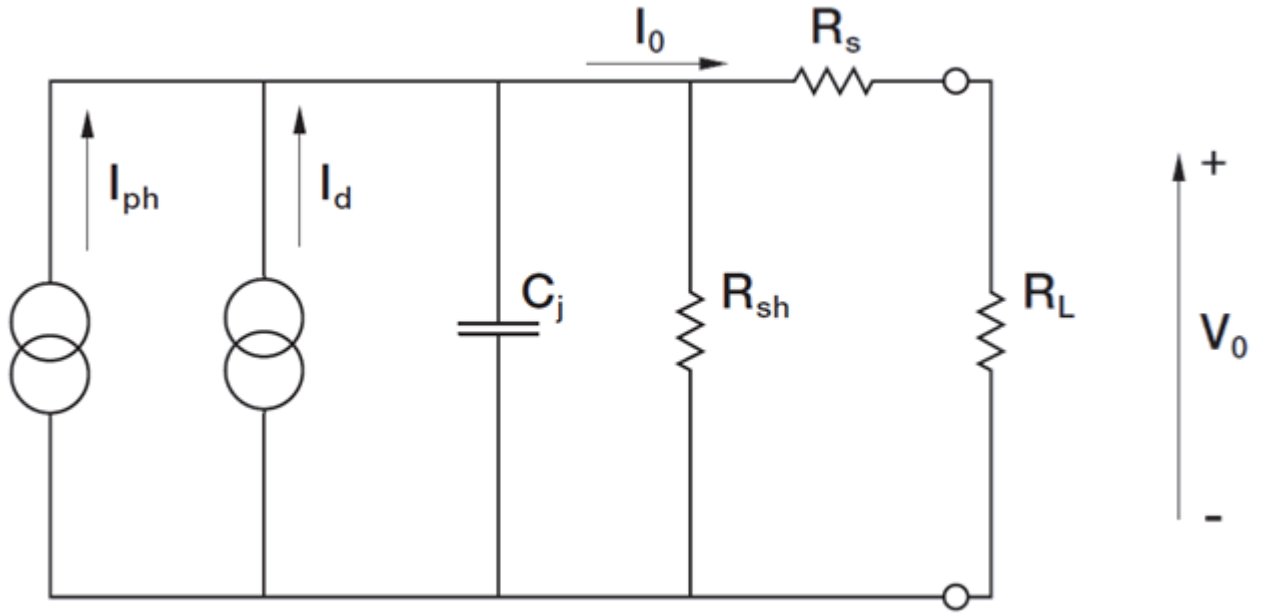
Photodiode Responsivity (sensitivity) =
ratio of the photocurrent I_{ph} to the incident light power P at a given wavelength: $R_\lambda = \frac{I_{ph}}{P}$



R_λ depends on:

- Incident light wavelength
- Applied V_{rev}
- Temperature





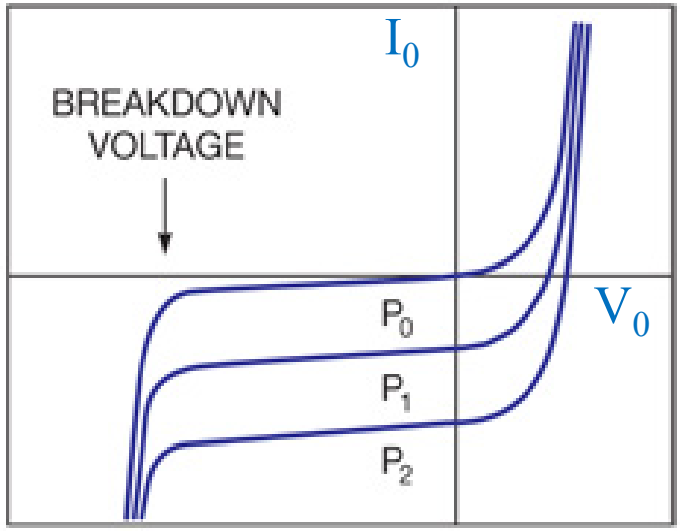
I_{ph} = photo- generated current

I_d = current of the ideal pn-junction (dark current)

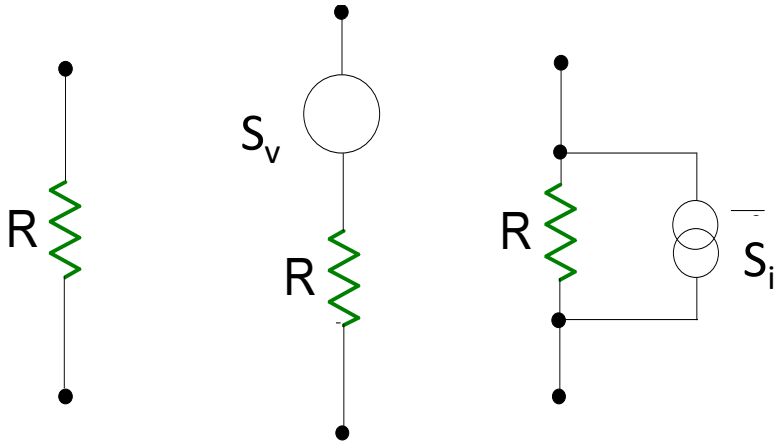
C_j = junction capacitance: $C_j = \frac{\varepsilon \cdot A}{W_d}$

R_{SH} = shunt resistance: slope at 0V

R_s = series resistance: $R_s = \frac{(W_s - W_d) \cdot \rho}{A} + R_c$



Thermal (Jhonson) noise → Brownian motion of charges



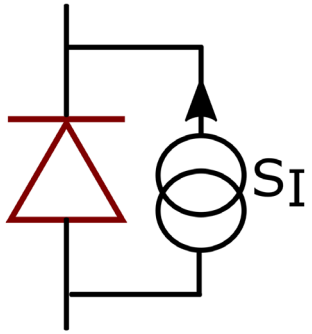
Spectral density

$$S_v = 4KT \cdot R \quad \left[\frac{V^2}{Hz}\right]$$

$$S_i = \frac{4KT}{R} \quad \left[\frac{A^2}{Hz}\right]$$

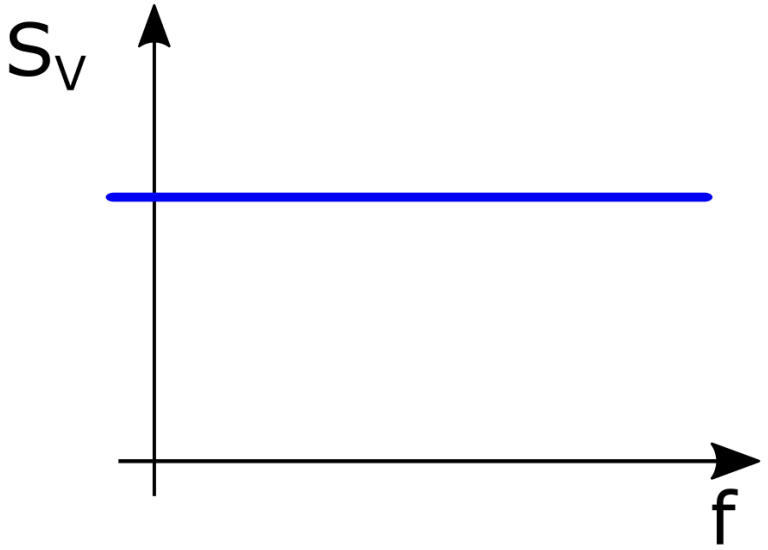
$$4kT = 1.66 \cdot 10^{-20} \frac{V^2}{Hz \cdot \Omega}$$

Shot noise → "granularity" of charge crossing a junction



Spectral density

$$S_i = 2q \cdot I$$

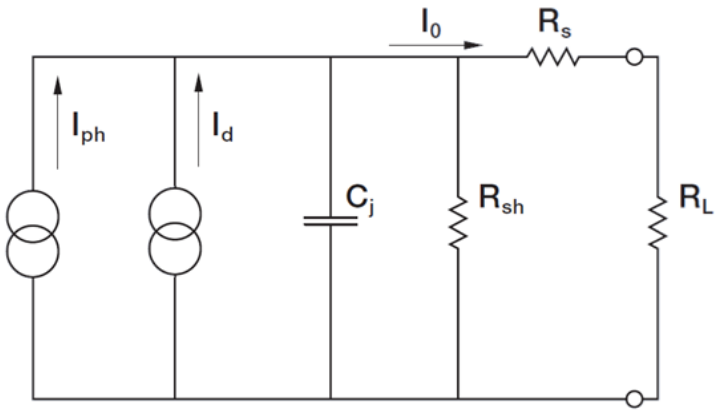


Noise standard deviation: $V_n = \sqrt{S_v \cdot \Delta f}$ and $I_n = \sqrt{S_i \cdot \Delta f}$

- **Shot noise**

  = statistical fluctuation in both the photocurrent and the dark current:

$$I_{sn} = \sqrt{2q(I_{ph} + I_d) \cdot \Delta f}$$



- **Thermal (Jhonson) noise**

  = thermal generation of the shunt resistance:

$$I_{jn_Rsh} = \sqrt{\frac{4kT\Delta f}{R_{sh}} \cdot \frac{R_{sh}^2}{(R_{sh} + R_s)^2}} \approx \sqrt{\frac{4kT\Delta f}{R_{sh}}}$$

  Thermal noise of the series resistance is negligible:

$$I_{jn_Rs} = \sqrt{\frac{4kT\Delta f}{R_s} \cdot \frac{R_s^2}{(R_{sh} + R_s)^2}} \approx \sqrt{\frac{4kT\Delta f}{R_{sh}} \cdot \frac{R_s}{R_{sh}}} = I_{jn_{Rsh}} \cdot \sqrt{\frac{R_s}{R_{sh}}}$$

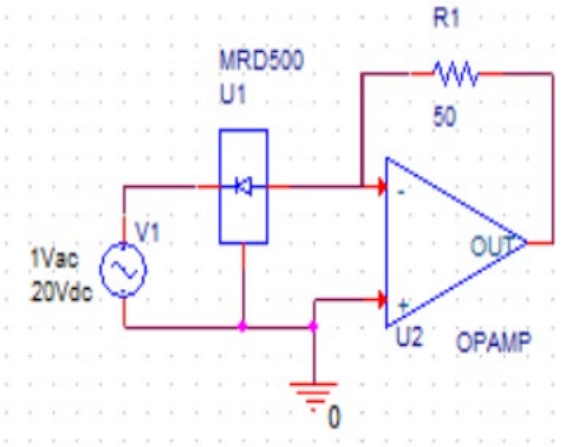
→ **Noise Equivalent Power (NEP)**

= amount of incident light power on a photodetector, which generates a photocurrent equal to the noise current.

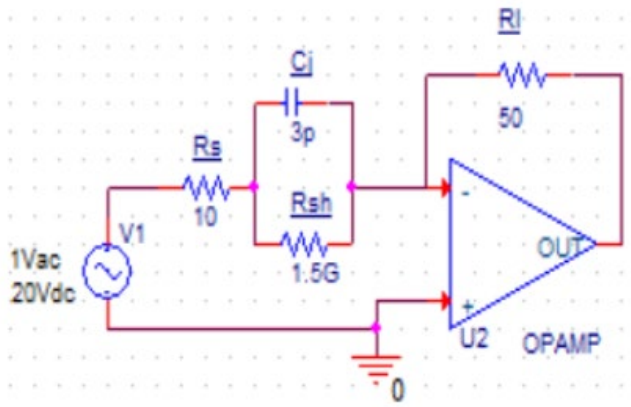
$$NEP = \frac{I_n}{R_\lambda}$$

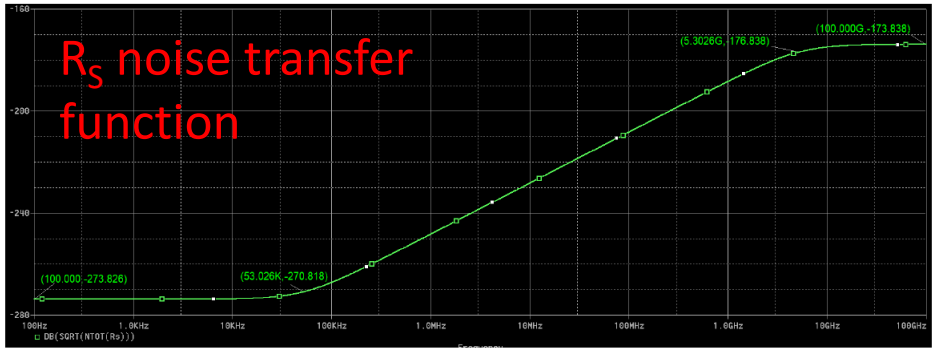
Noise SPICE simulations

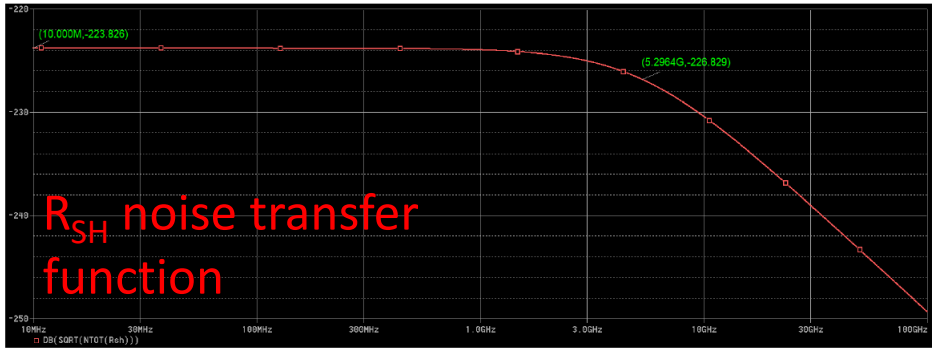
Commercial diode model



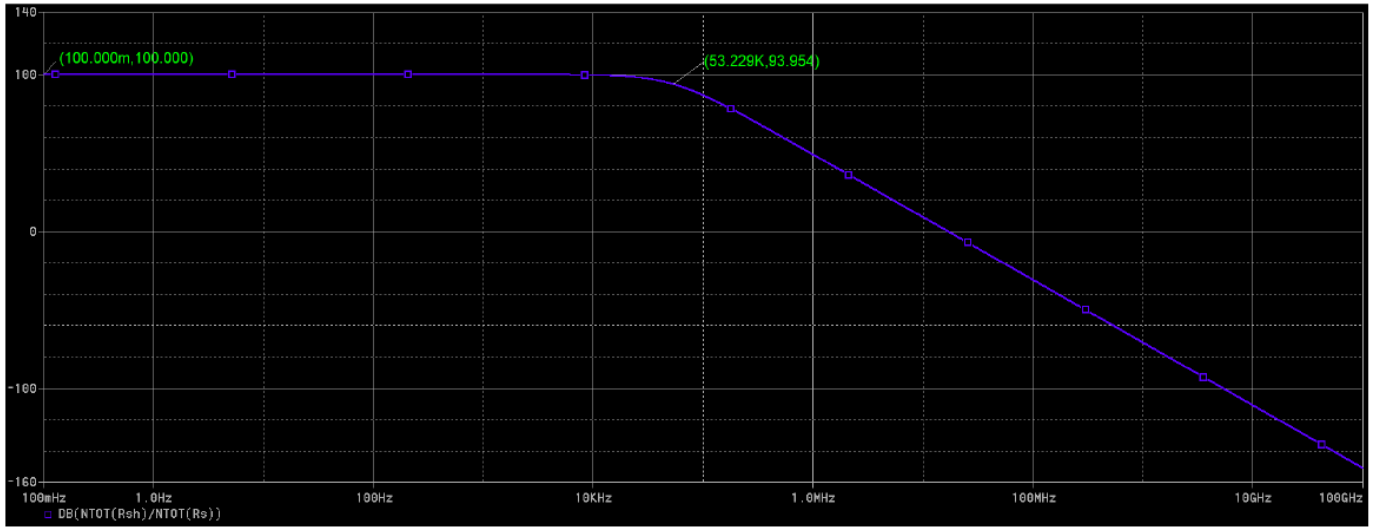
Diode modelled with passive components

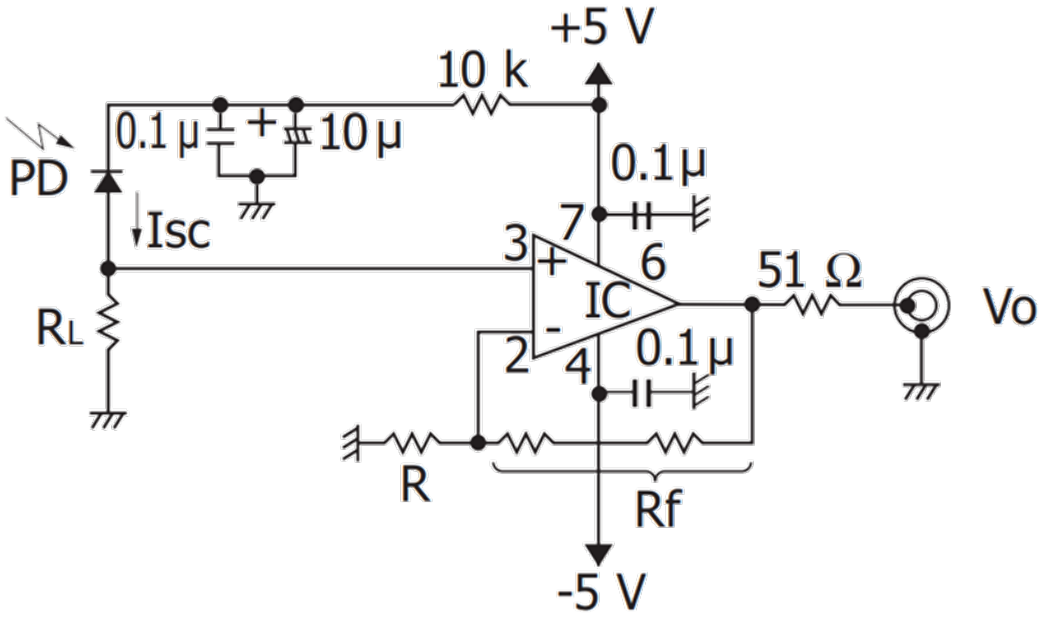






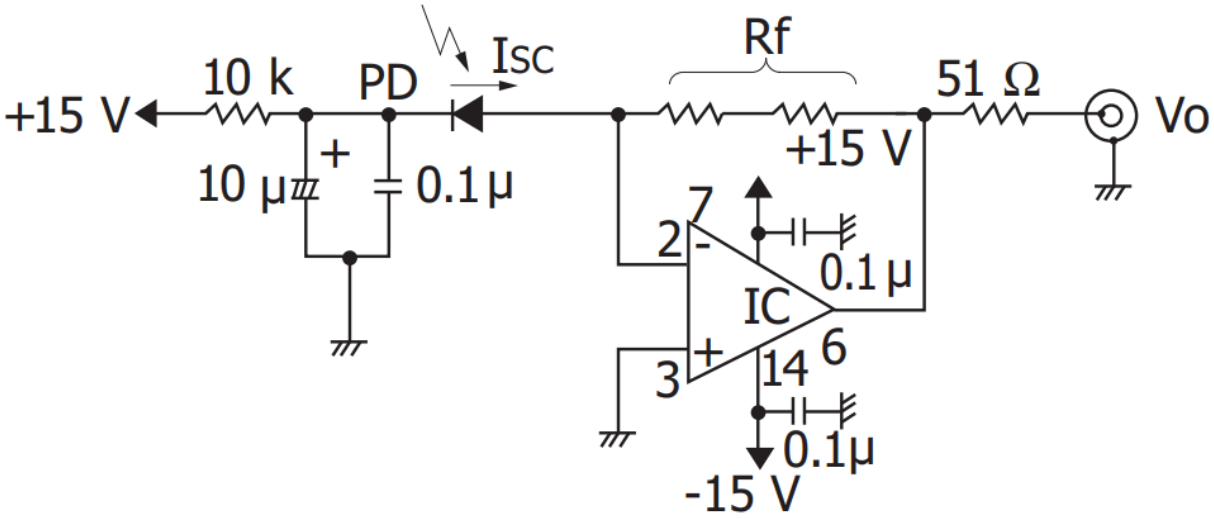
Ratio between R_{SH} and R_S noise transfer functions





Non-inverting configuration

$$V_o = I_{SC} \cdot R_L \cdot \left(1 + \frac{R_f}{R}\right)$$

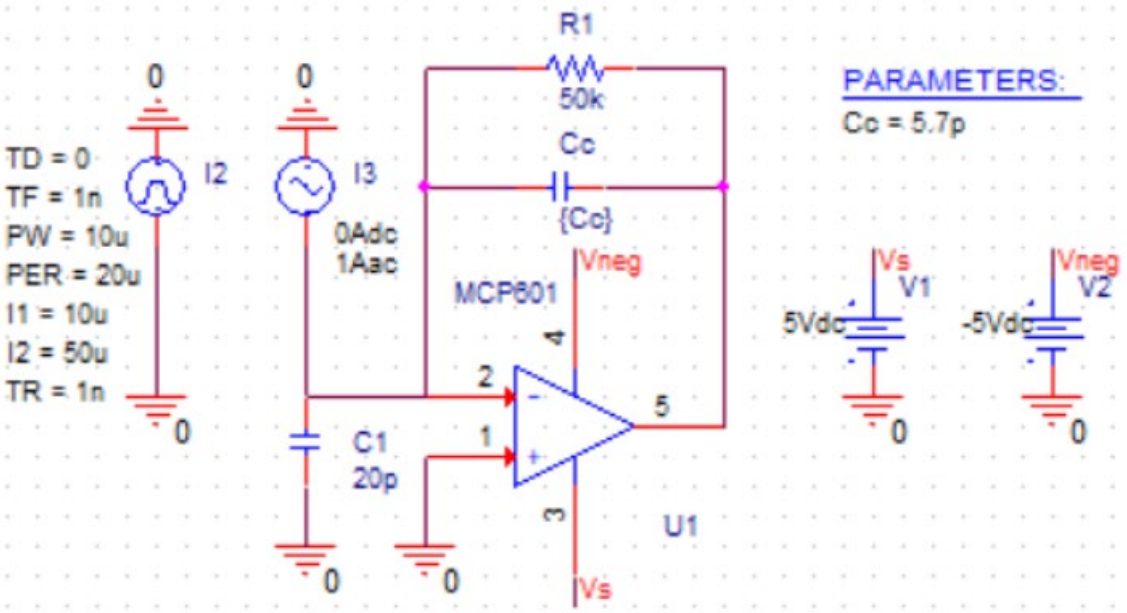


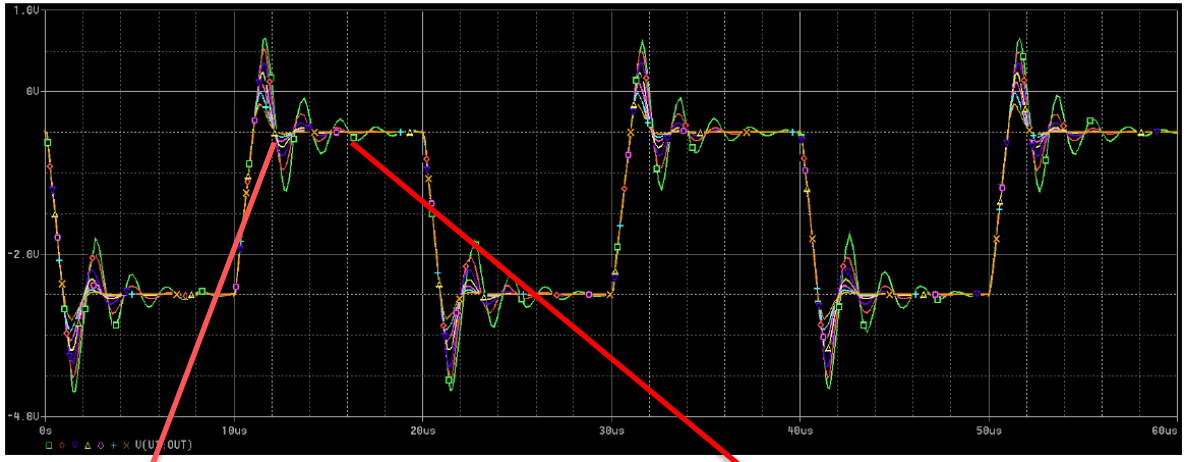
Transimpedance Amplifier

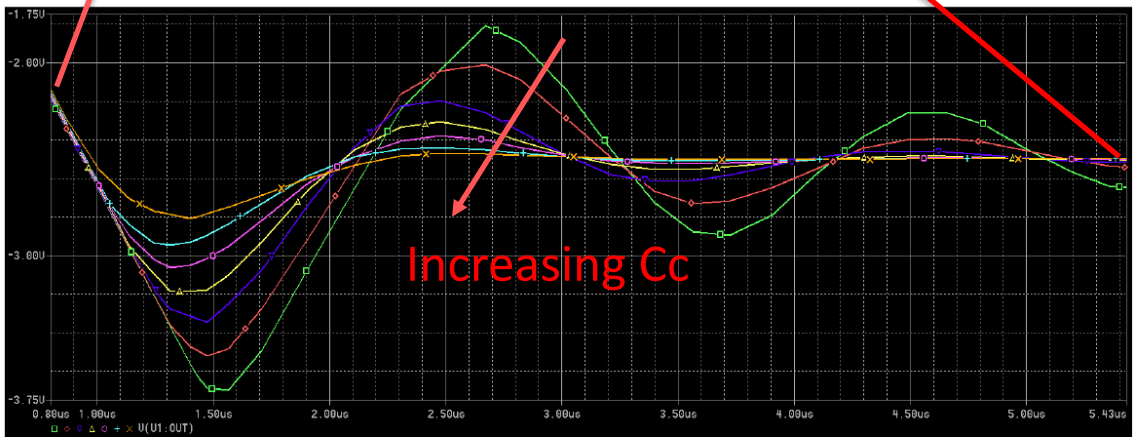
$$V_o = I_{SC} \cdot R_f$$

Typically, must be compensated!

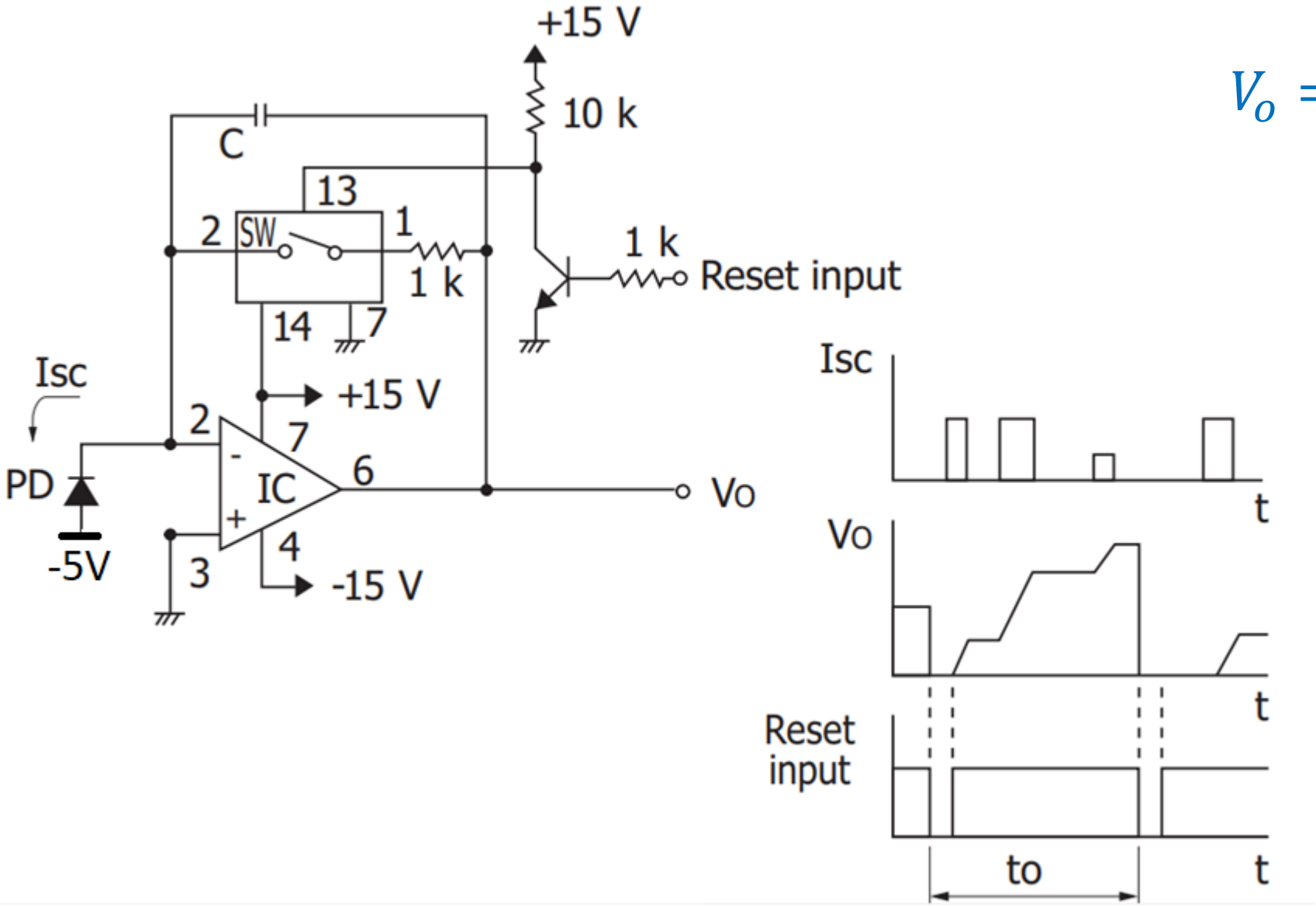
TIA instability







$$V_o = \frac{\overline{I_{SC}} \cdot t_o}{C}$$



Photodiode applications

Camera
- Light Meters
- Automatic Shutter Control
- Auto-focus
- Photographic Flash Control

Medical
- X ray Detection
- Pulse Oximeters
- Blood Particle Analyzers

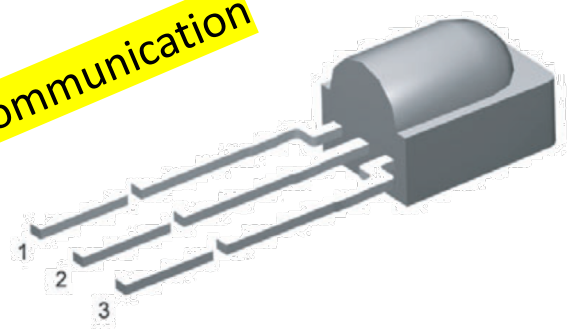
Safety Equipment
- Smoke Detectors
- Flame Monitors
- Security Inspection Equipment
- Intruder Alert - Security System

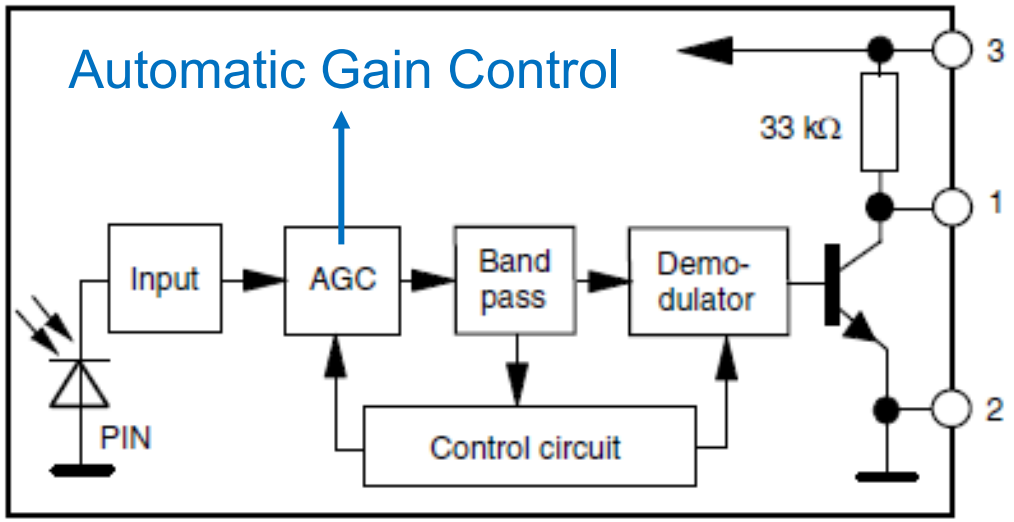
Automotive
- Twilight Detectors
- Climate Control - Sunlight Detector

Industry
- Bar Code Scanners
- Light Pens
- Brightness Controls
- Rotary Encoders
- Position Sensors
- Surveying Instruments
- Copiers - Density of Toner

Communications
- Fiber Optic Links
- Optical Communications

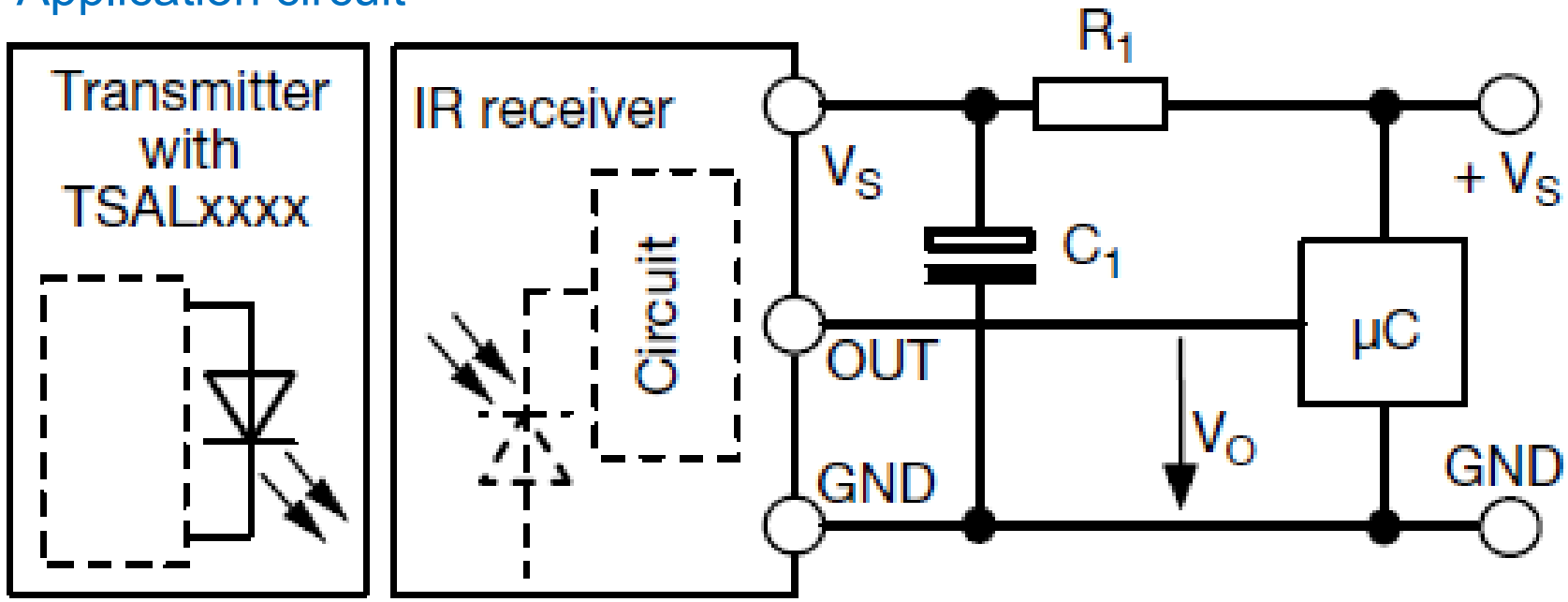
LAB 11: IR communication



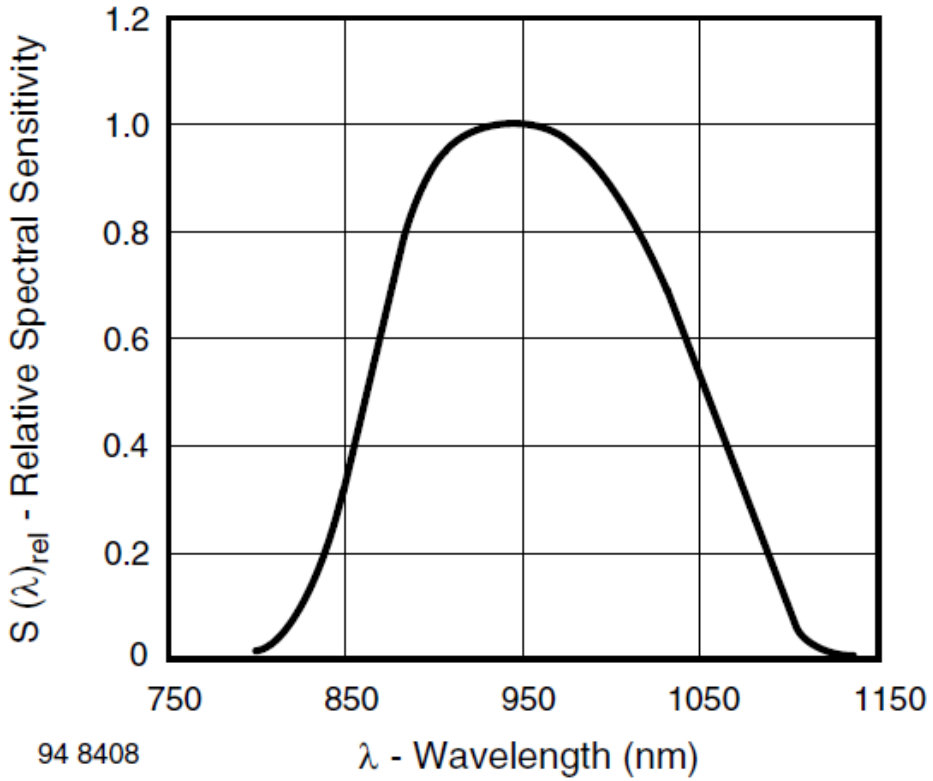


Carrier frequency = 38 kHz

Application circuit

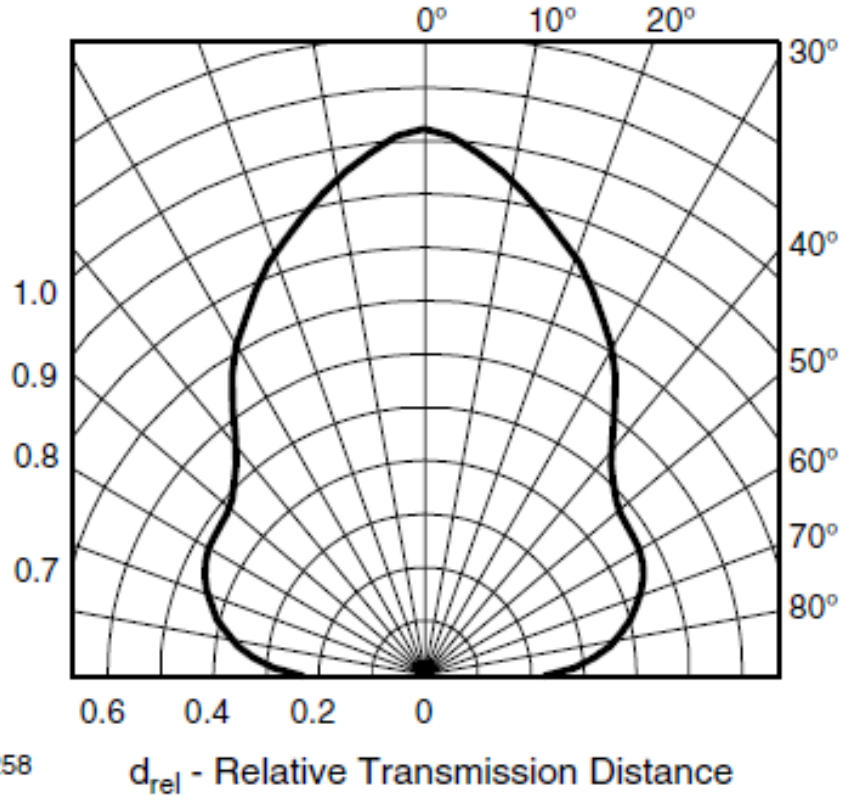


Relative spectral sensitivity



The peak sensitivity is at 950 nm
→ Near Infra-Red (NIR)

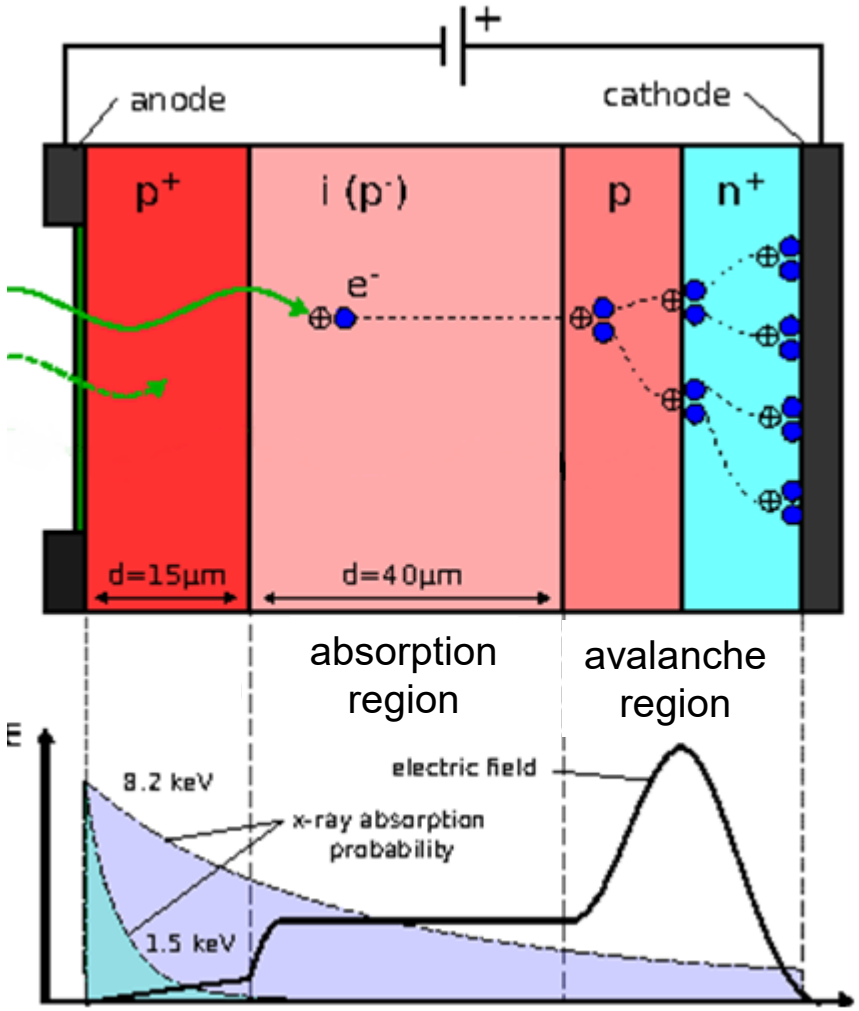
Relative transmission distance



The sensitivity for light beam
with 0° angle is the highest

The same sensitivity for a light beam
with 80° angle is obtain at 0.35 relative distance.

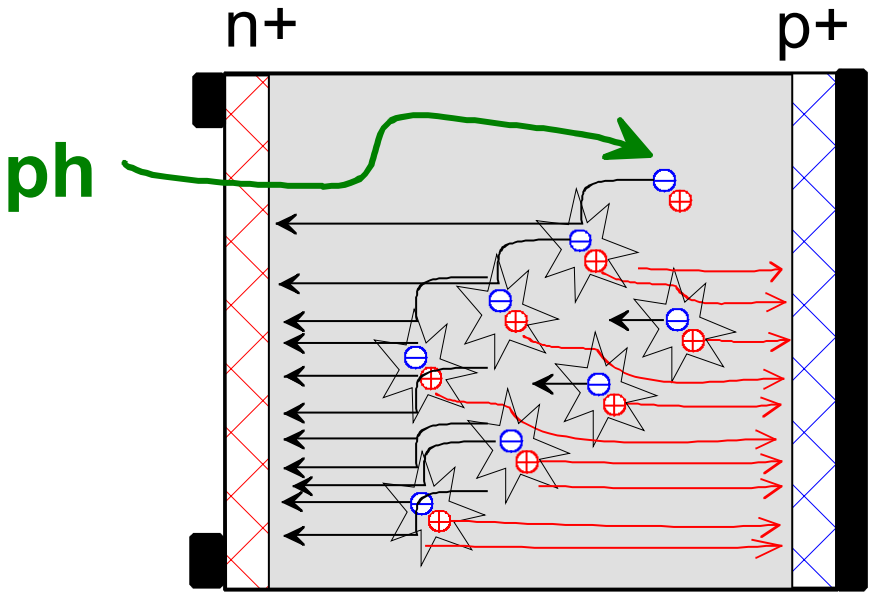
Avalanche Photodiode (APD)

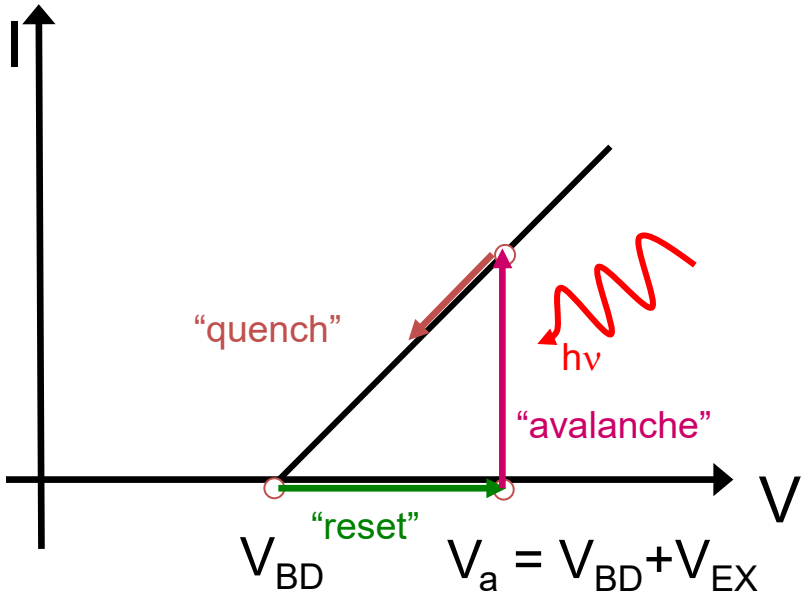


- P-i-N junction
- Bias: below but close to breakdown
- Analog output proportional to incoming light
- Internal multiplication (gain: M ≈ 100)

Basic structure:

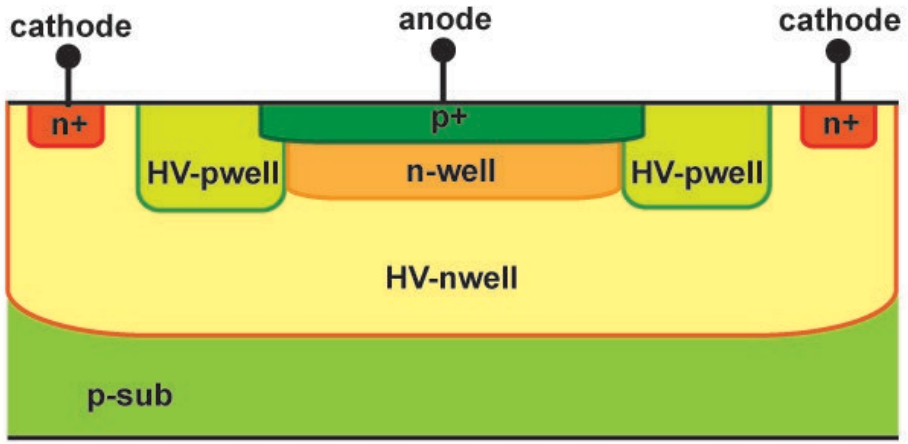
- absorption region = separate photo-generated holes and electrons
- multiplication region = high electric field to provide internal photo-current gain by impact ionization

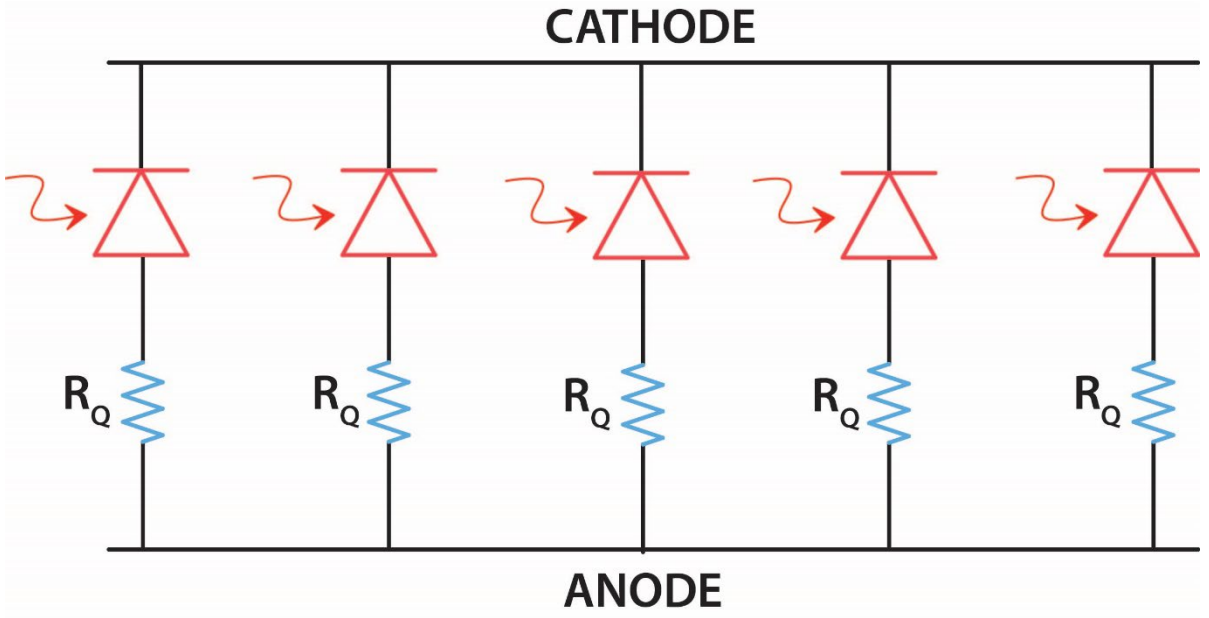




- P-N junction
- Bias: well *ABOVE* breakdown
- **digital output**
- single photon sensitivity
- no read-out noise
- output: >10 mA current for every photon

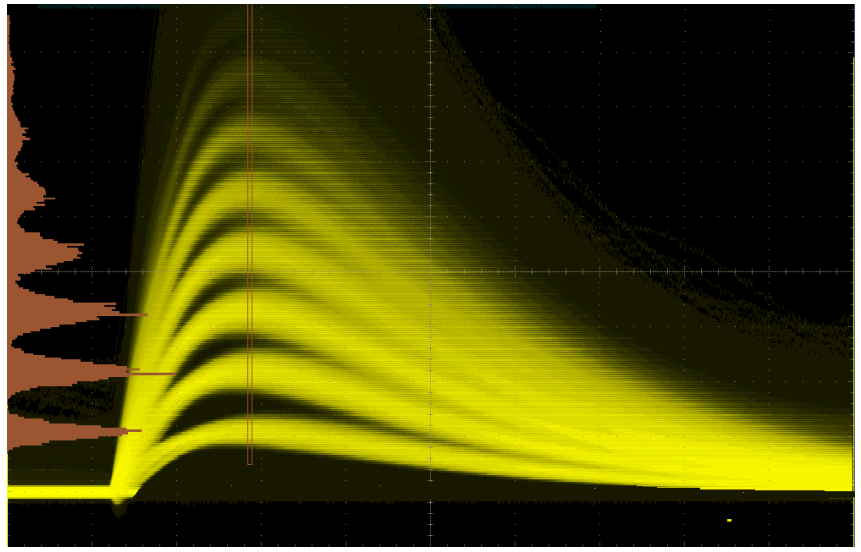
solid-state device:
can be integrated together
with electronics
(counter & TDC)





Hundreds / thousands **SPADs**
+ quenching resistance in parallel

- ✓ Single photon sensitivity
- ✓ Photon number resolving capability
- ✓ Large active area



In the dark → R very high (up to 1MΩ)
Exposed to light → R drops dramatically (few ohms)

LAB 6: LDR and ADC

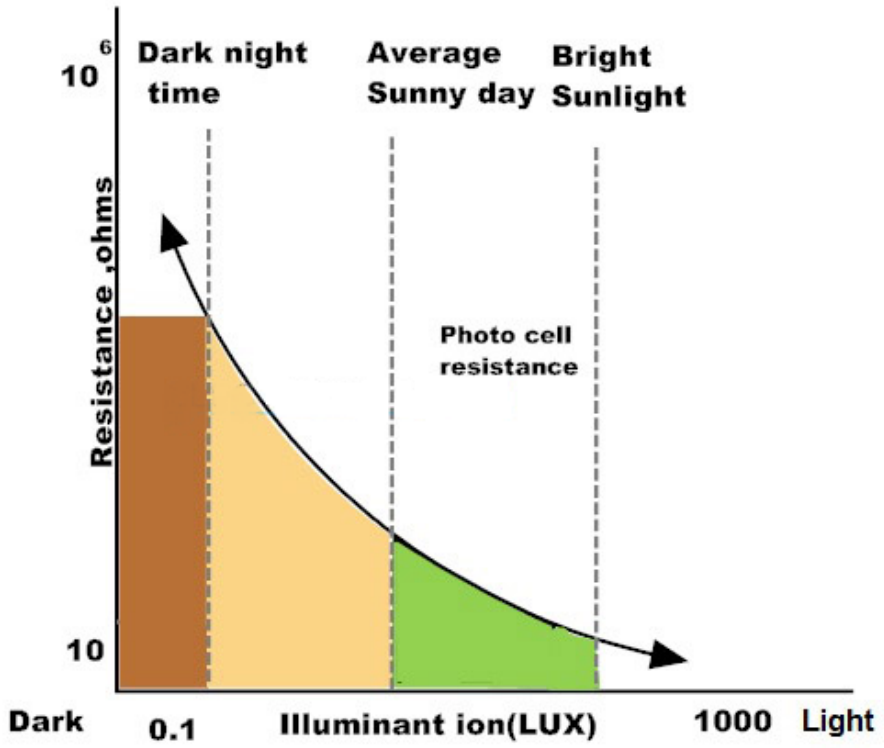
Working principle:

Photons excite electrons from the valence band to the conduction band

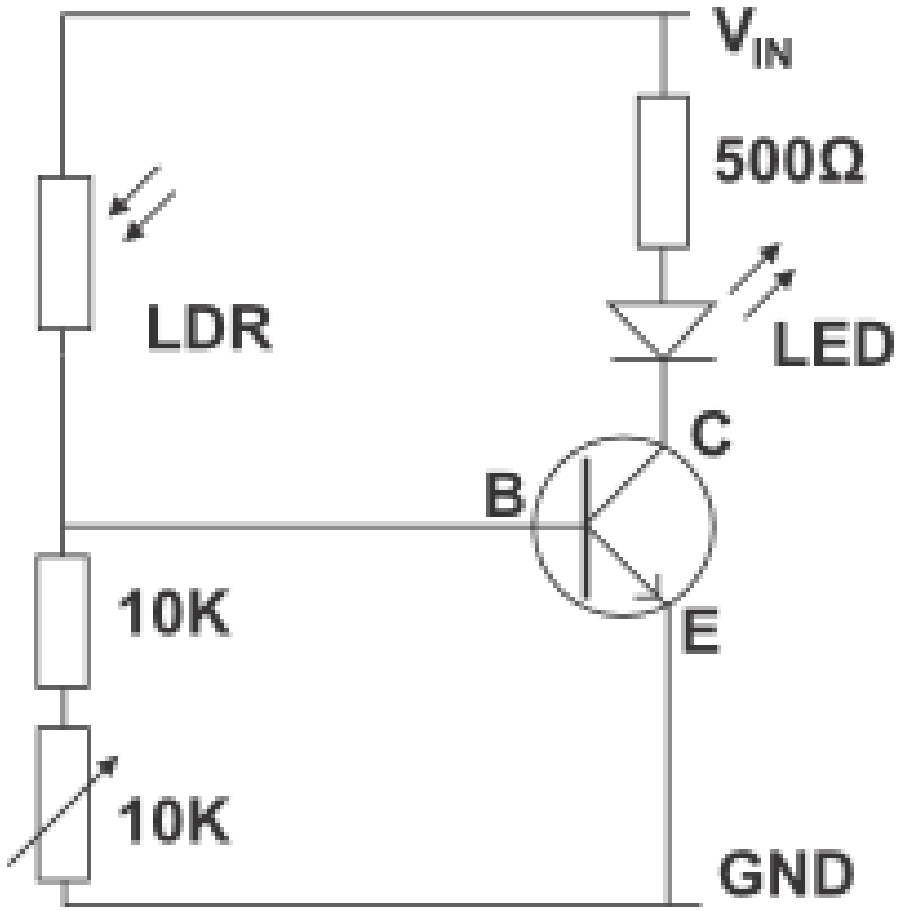
→ more free electrons in the material

→ lower resistance

But…
- low sensitivity
- non-linear characteristic
- sensitive to temperature changes







WITHOUT LIGHT

High LDR

V_{BE} < 0.7 → BJT off

→ LED off

WITH LIGHT

Low LDR

V_{BE} = 0.7 → BJT on

→ LED on